

Collaborators:

Problem Set 2: Q27.3; P27.17+; P27.4; P27.5; Purcell 1.71; *E27.25

Q27.3: (5 points) By analogy with Φ_E , how would you define the flux Φ_E of a gravitational field? What is the flux of the Earth's gravitational field through the boundaries of a room, assumed to contain no matter? Through a spherical surface closely surrounding the earth? Through a spherical surface with radius just within the Moon's orbit?

P27.17+: (5 points) A solid nonconducting sphere of radius R carries a nonuniform charge distribution, with charge density $\rho = \rho_s \frac{r}{R}$ where ρ_s is a constant and r is the distance from the center of the sphere. Show that

(a) the the total charge on the sphere is $Q = \pi\rho_s R^3$

(b) the magnitude of the electric field inside the sphere is given by

$$E(r) = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^4} r^2$$

(c) Write down the electric field outside the sphere, $r > R$. You do not need to derive your result, but explain how you know it.

P27.4: (5 points) Figure 27-33 shows a charge $+q$ arranged as a uniform conducting sphere of radius a and placed at the center of a spherical conducting shell of inner radius b and outer radius c . The outer shell carries a charge of $-q$. Find $E(r)$ at the following locations:

- (a) within the sphere ($r < a$)
- (b) between the sphere and the shell ($a < r < b$)
- (c) inside the thickness of the shell ($b < r < c$)
- (d) outside the shell ($r > c$)
- (e) What charges appear on the inner and outer surfaces of the shell?

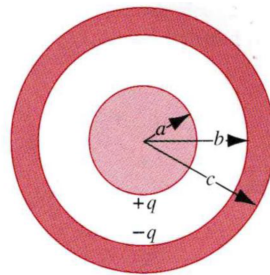
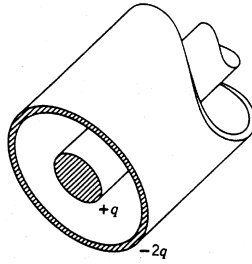


FIGURE 27-33. Problem 4.

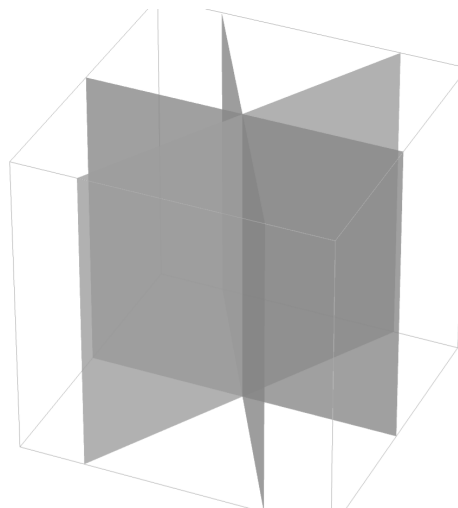
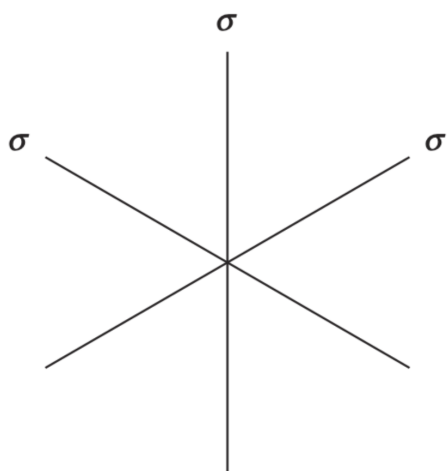
P27.5: (5 points) A very long conducting cylinder (length L) carrying a total charge $+q$ is surrounded by a conducting cylindrical shell (also of length L) with total charge $-2q$, as shown in cross section in Fig. 27.34. Use Gauss's law to find

1. the electric field at points outside the conducting shell
2. the distribution of the charge on the conducting shell
3. the electric field in the region between the cylinders



Purcell 1.71: (5 points) The figures below shows three infinite sheets of charge, all with the same surface charge density σ , intersecting at equal angles – the first image is a top-down view and the second is a 3D depiction of the planes.

- (a) Use superposition to find the electric field, in vector form, at all points in space.
- (b) Use Gauss's Law to find the electric field.
- (c) What is the electric field in the analogous setup where there are N sheets instead of three?



***E27.25:** (10 points) Charge is distributed uniformly throughout an infinitely long cylinder of radius R .

- (a) Show that within the cylinder ($r < R$) the electric field magnitude $E(r)$ at a distance r from the cylinder axis is given by

$$E(r) = \frac{\rho r}{2\epsilon_0}$$

- (b) What is the electric field magnitude $E(r)$ for $r > R$?